Fields

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1 Motivation

So far we worked technically at a *single point* p of a manifold: one tangent space T_pM, a vector in it, a covector in T_p^*M, and a basis induced by a chart. As physicists we want *fields*: at every point of the manifold a vector (or covector, or tensor). The usual way to introduce this is "just imagine a vector at every point", but imagining is dangerous (you may imagine something different from someone else), and it does not say what it means for the assignment to be *smooth*. The proper tool is the theory of **bundles**. (Fields were introduced into physics by Faraday, who knew no mathematics by modern standards; pictures suffice for intuition, but each picture below is backed by a precise definition.)

2 Bundles

Definition 1 (Bundle)**.** A *bundle* is a triple (E, π, M) consisting of

- a smooth manifold E, the *total space*,
- a smooth manifold M, the *base space*,
- a *surjective smooth* map $\pi : E \to M$, the *projection map*.

(Some authors say "smooth bundle"; we only consider these.) Any such triple is a bundle, so there are enormously many.

Example 1. Let E be a two-dimensional cylinder and M a circle, each with its topology and smooth atlas. One choice of π sends each point of the cylinder to the point of the circle directly below it, so that all points on a vertical line project to the same base point. Any other smooth surjective map would do equally well to make (E, π, M) a bundle.

Definition 2 (Fiber)**.** For a bundle (E, π, M) and $p \in M$, the *fiber over* p is

$$\mathrm{preim}_\pi(\{p\}) = \{e \in E \mid \pi(e) = p\}.$$

The projection is the primary datum; the fibers are derived objects. In the cylinder example the fiber over a point of the circle is the vertical line above it.

Definition 3 (Section)**.** A *section* of a bundle (E, π, M) is a map $\sigma : M \to E$ such that

$$\pi \circ \sigma = \mathrm{id}_M.$$

A *smooth* section is a section that is a smooth map.

Without the condition, σ could send a base point p to a point lying in a different fiber. The condition says: going up with σ and projecting down returns the point you started from, i.e. $\sigma(p)$ lies in the fiber over p. A section needs the whole bundle structure (E, M and π).

Sections are the fields. Which kind of field a section describes depends on the fibers:

- fibers = tangent spaces ⇒ sections are *vector fields*;
- fibers = cotangent spaces ⇒ *covector fields*;
- fibers = tensor spaces ⇒ *tensor fields*.

In each case a field associates to each point an object in *that point's own* fiber, and not a vector from somewhere else.

Remark (Aside: quantum mechanics). In quantum mechanics the wave function is properly a section of a complex line bundle ($\mathbb{C}$-fibers) over position space, not simply a function. Locally a section can be represented by a function, but globally, and for the question of what the momentum operator is (a partial derivative, or a covariant derivative with a suitable connection, which matters e.g. on a sphere), the distinction matters. Covariant derivatives are introduced later in the course. The lecturer deliberately did not go into detail; it is only meant to show that the field concept also appears in quantum mechanics.

3 The tangent bundle

Vector fields are sections of the tangent bundle, so we construct it for an arbitrary smooth manifold $(M, \mathcal{O}, \mathcal{A})$, taken as the base space. This is also good practice with all the previous concepts. We have to supply a total space that is a *smooth manifold*, and a smooth surjective projection.

3.1 Step 1: the set TM and the projection

Definition 4 (Tangent bundle, as a set)**.**

$$TM := \bigsqcup_{p \in M} T_pM,$$

the *disjoint* union of all tangent spaces.

The disjoint union emphasizes that elements of different tangent spaces are never identified: a vector in T_pM is a different object from a vector in T_qM, even if one throws them all into one bag. (The lecturer's analogy: everybody in the room has a nose; collecting all the noses in a bag does not make one person's nose another's.)

Define $\pi : TM \to M$ by

$$\pi(X) = \text{the unique } p \in M \text{ with } X \in T_pM.$$

Uniqueness comes from the disjointness of the union, and π is surjective since every point's tangent space was included. At this stage TM is a mere set, so we cannot yet ask whether π is smooth.

3.2 Step 2: a topology on TM

We need a topology first (a smooth manifold is in particular a topological manifold). We want π to be continuous, otherwise it cannot be smooth. The discrete topology would make π continuous trivially, but would ignore the data in π. It is better to choose the *coarsest* topology that makes π continuous, the so-called *initial topology* with respect to π:

$$\mathcal{O}_{TM} := \big\{ \mathrm{preim}_\pi(U) \;\big|\; U \in \mathcal{O} \big\}.$$

The pre-images of open sets must be open for π to be continuous, so these are needed; and nothing more is needed. That $\mathcal{O}_{TM}$ is indeed a topology (contains $\emptyset$ and TM, closed under finite intersections and arbitrary unions) is an exercise left for the tutorial. (The initial topology is easy to write down; the analogous *final* topology, where the topology is known on the target and wanted on the domain, would need the notion of a topological basis and is trickier.)

3.3 Step 3: a smooth atlas on TM

We construct charts on TM from the charts of M and read off that TM is a topological manifold and in fact a smooth manifold. Let $(U, x) \in \mathcal{A}$ with $\dim M = d$. The chart domain is

$$TU = \mathrm{preim}_\pi(U) = \bigsqcup_{p \in U} T_pM,$$

which is open in TM by the definition of $\mathcal{O}_{TM}$.

Definition 5 (Induced chart)**.** Define $\xi_x : TU \to \mathbb{R}^{2d}$ by

$$\xi_x(X) = \Big(x^1\big(\pi(X)\big),\ \ldots,\ x^d\big(\pi(X)\big),\ (\mathrm{d}x^1)_{\pi(X)}(X),\ \ldots,\ (\mathrm{d}x^d)_{\pi(X)}(X)\Big).$$

The first d entries are the coordinates of the base point $\pi(X)$ in the chart x. The last d entries are the components of X with respect to the chart-induced basis $\partial/\partial x^i|_{\pi(X)}$ of $T_{\pi(X)}M$: indeed if $X = X^i\,\partial/\partial x^i|_{\pi(X)}$ then, since $(\mathrm{d}x^j)_p$ is the dual basis (Lecture 5), $(\mathrm{d}x^j)_{\pi(X)}(X) = X^j$. So ξ_x records both where the vector sits and what its components are. The dimension doubles, $\dim TM = 2d$, as expected (each tangent space has dimension d).

Inverse. $\xi_x^{-1} : \xi_x(TU) \to TU$ sends $(\alpha^1, \ldots, \alpha^d, \beta^1, \ldots, \beta^d)$ to

$$\beta^i \left.\frac{\partial}{\partial x^i}\right|_{x^{-1}(\alpha)}.$$

The α's are the coordinates of the base point $p = x^{-1}(\alpha)$ and the β's are components with respect to the basis at p, so the vector (including which tangent space it lives in) is fully reconstructed.

Transition maps. Let $(V, y) \in \mathcal{A}$ be another chart with $U \cap V \neq \emptyset$. Then

$$\xi_y \circ \xi_x^{-1} : (\alpha, \beta) \longmapsto \Big(y^i \circ x^{-1}(\alpha)\ \ (i = 1, \ldots, d),\ \ \beta^m \left.\frac{\partial y^i}{\partial x^m}\right|_{x^{-1}(\alpha)}\ \ (i = 1, \ldots, d)\Big).$$

To see this, apply ξ_y to $\beta^m\,\partial/\partial x^m|_{x^{-1}(\alpha)}$: the first d entries are $y^i(x^{-1}(\alpha))$, the base point coordinates in the chart y; the last d entries are

$$(\mathrm{d}y^i)\Big(\beta^m \left.\tfrac{\partial}{\partial x^m}\right|_{x^{-1}(\alpha)}\Big) = \beta^m \left.\frac{\partial}{\partial x^m}\right|_{x^{-1}(\alpha)} y^i = \beta^m\,\partial_m\big(y^i \circ x^{-1}\big)(\alpha),$$

using that $\mathrm{d}y^i$ is linear and that the gradient acts as $(\mathrm{d}f)_p(X) = X(f)$. The first block is the chart transition map on M; the second block is exactly the transformation law for vector components from Lecture 5.

Smoothness. The first d entries are smooth in α because the transition maps on M are C^∞. The last d entries are linear in β and have coefficients $\partial_m(y^i \circ x^{-1})(\alpha)$, i.e. first derivatives of C^∞ maps, which are again C^∞. Hence the transition maps are C^∞ and

$$\mathcal{A}_{TM} = \big\{(TU, \xi_x) \;\big|\; (U, x) \in \mathcal{A}\big\}$$

is a C^∞ atlas on TM. (The lecture does not spell out the check that the ξ_x are homeomorphisms onto open subsets of $\mathbb{R}^{2d}$ for the topology $\mathcal{O}_{TM}$; the atlas is said to exhibit TM as a topological and then smooth manifold.)

The projection is smooth. In the charts ξ_x on TM and x on M, π reads $(\alpha, \beta) \mapsto \alpha$, a projection onto the first d components, which is smooth.

Theorem 1. *For any smooth manifold $(M, \mathcal{O}, \mathcal{A})$, the triple (TM, π, M) is a bundle in the sense above, called the* tangent bundle *of M.*

Along the way we also obtained an explicit atlas, so we can work in charts on TM if we want to.

Example 2. For $M = S^1$, TS^1 is, as a set, a cylinder: over each point of the circle sits the tangent line at that point.

4 Vector fields

All this effort is in order to be able to write the following definition.

Definition 6 (Smooth vector field)**.** A *smooth vector field* on M is a smooth section $\psi : M \to TM$ of the tangent bundle, i.e. a smooth map with

$$\pi \circ \psi = \mathrm{id}_M.$$

This makes precise the intuition "a vector at each point", and also what it means for the association to be smooth: ψ is a smooth map between smooth manifolds. It required TM to be a smooth manifold. Without that one could still speak of non-smooth vector fields (sections), but not of smooth ones.

Example 3. On S^1, draw a vector field that points forward at some places, shrinks, vanishes, reverses, vanishes again and grows again. As a section of the cylinder TS^1 it picks, over each base point, one point of the vertical line over it: the zero vector where the field vanishes, a point in the negative direction where it points backward, and so on. The whole section is a curve winding around the cylinder once.

The set of all smooth vector fields is written $\Gamma(TM)$.

4.1 Evaluation and action on functions

For $\psi \in \Gamma(TM)$ and $p \in M$ write $\psi_p := \psi(p) \in T_pM$. A vector field acts on a smooth function $f \in C^\infty(M)$ to give a smooth function $\psi(f) : M \to \mathbb{R}$ by

$$\big(\psi(f)\big)(p) := \psi_p(f),$$

i.e. at each point the vector ψ_p acts on f as before. This is a new kind of application rule (the evaluation point "jumps in between"), but it is the standard notation: a vector field acting on a function gives a function.

4.2 $\Gamma(TM)$ is a $C^\infty(M)$-module, not a vector space

The set $C^\infty(M)$ of smooth functions is not just a real vector space under pointwise operations; it also has a pointwise multiplication, which is commutative and associative, with neutral element 1 (the constant function) and with distributivity. Addition has the neutral element 0 (the zero function) and inverses $-f$. But *multiplicative inverses fail*: a function that is not identically zero may vanish at some points, and then has no inverse. So $C^\infty(M)$ is a commutative ring with unit, but *not a field* (unlike $\mathbb{R}$ or $\mathbb{Q}$; $\mathbb{Z}$ is not a field either).

$\Gamma(TM)$ carries:

- an addition: $(\psi + \phi)(f) := \psi(f) + \phi(f)$, adding the two resulting functions;

- a multiplication by smooth *functions*: for $g \in C^\infty(M)$, $(g\psi)_p := g(p)\,\psi_p$, i.e. scaling the vector field differently at each point.

Restricting to constant g would only give an $\mathbb{R}$-vector space; allowing arbitrary functions gives the vector space axioms over the ring $C^\infty(M)$:

Theorem 2 (Upshot)**.** $\Gamma(TM)$ *is a* $C^\infty(M)$-module*: it satisfies the vector space axioms, but with scalars from the ring* $C^\infty(M)$ *instead of a field.*

Mathematicians give this its own name because many of the theorems for vector spaces fail for modules. In particular:

- every vector space has a basis (the proof needs the axiom of choice);
- **no such result holds for modules**: a module need not have a basis.

4.3 Consequence: no global basis of vector fields

If $\Gamma(TM)$ had a basis $\chi_1, \dots, \chi_d$, every vector field could be written globally as $\psi = f^i \chi_i$ with component *functions* $f^i \in C^\infty(M)$. This does not hold in general, and the standard counterexample is the sphere:

Example 4 (The sphere cannot be combed)**.** Every smooth vector field on S^2 must vanish somewhere ("a hairy sphere cannot be combed smoothly", the hairy ball theorem; the lecturer mentioned Morse theory as a route to a proof). A basis of the module would need χ_1, χ_2 spanning T_pS^2 at *every* point p (otherwise a field with a component in the missing direction there could not be expressed), so neither could vanish anywhere. Hence $\Gamma(TS^2)$ has no basis.

$\Gamma(TM)$ may well have a basis for some manifolds (e.g. $\mathbb{R}^2$), but this is not guaranteed, and already the sphere fails. So it is *not* an option to "just expand everything in a global basis". What we can do:

- expand locally on chart domains: for a chart (U, x) the maps

$$\frac{\partial}{\partial x^i} : U \to TU, \qquad p \mapsto \left.\frac{\partial}{\partial x^i}\right|_p$$

  are smooth vector fields on U forming a basis there (the chart-induced basis fields). They are defined only on the chart domain; this is not in contradiction with the above, and charts fail at similar points (at a point where a chart-induced field degenerates the chart cannot reproduce a component in that direction);
- work basis-free wherever possible.

This is yet another blow to the idea of "just doing everything in charts": vector fields can locally, but not globally, be handled by components.

5 Tensor fields

Almost everything in general relativity that lives on the manifold (matter, and also the geometry itself) is described by a tensor field.

Cotangent bundle. Exactly as for TM one constructs the cotangent bundle T^*M (as a disjoint union of the T^*_pM with the initial topology, charts mapping a covector to the coordinates of its base point and its components with respect to the basis $(\mathrm{d}x^i)_p$). This is an exercise in the tutorial. Its smooth sections are the *covector fields* $\Gamma(T^*M)$, again a $C^\infty(M)$-module.

Definition 7 (Tensor field)**.** An (r,s) *tensor field* T is a $C^\infty(M)$-multilinear map

$$T : \underbrace{\Gamma(T^*M) \times \cdots \times \Gamma(T^*M)}_{r} \times \underbrace{\Gamma(TM) \times \cdots \times \Gamma(TM)}_{s} \longrightarrow C^\infty(M).$$

So T eats r covector fields and s vector fields and gives a smooth function (a real number at each point). "Multilinear" means: additive in each slot, and scaling a slot by a smooth function g scales the result by g. Linearity with respect to *real* scalars alone is not enough.

Warning. Many objects one writes down (such as the Riemann and torsion tensors, defined later) are multilinear when fed vector fields, in the sense of real scalars, but are not $C^\infty(M)$-linear. Only the $C^\infty(M)$-multilinear ones qualify as tensor fields, so it is a practical check to carry out every time.

Example 5 (Gradient of a function)**.** For $f \in C^\infty(M)$ define $\mathrm{d}f : \Gamma(TM) \to C^\infty(M)$ by

$$(\mathrm{d}f)(\psi) := \psi(f), \qquad \text{i.e.} \quad \big((\mathrm{d}f)(\psi)\big)(p) = \psi_p(f) = (\mathrm{d}f)_p(\psi_p).$$

It has the structure of a $(0,1)$ tensor field. It is additive in ψ, and it is $C^\infty(M)$-linear: for $g \in C^\infty(M)$,

$$(\mathrm{d}f)(g\psi) = (g\psi)(f) = g\,\psi(f) = g\,(\mathrm{d}f)(\psi),$$

since $(g\psi)_p(f) = g(p)\,\psi_p(f)$. Hence $\mathrm{d}f$ is a $(0,1)$ tensor field, i.e. a covector field (the *gradient* of f).

Summary

- A **bundle** is a triple (E,π,M) with E, M smooth manifolds and $\pi : E \to M$ smooth and surjective. The **fiber** over p is $\mathrm{preim}_\pi(\{p\})$. A **section** is a map $\sigma : M \to E$ with $\pi \circ \sigma = \mathrm{id}_M$; **sections are fields**, and the type of field is fixed by the fibers.
- The **tangent bundle** $TM = \bigsqcup_p T_pM$ with $\pi(X) =$ base point. Its topology is the initial topology $\{\mathrm{preim}_\pi(U)\}$, and its smooth atlas has charts $\xi_x(X) = (x^i(\pi(X)),\ (\mathrm{d}x^i)_{\pi(X)}(X))$ whose transition maps are the base transition map together with the Jacobian acting on components. TM is a smooth $2d$-manifold and π is smooth.
- A **smooth vector field** is a smooth section of TM. The set $\Gamma(TM)$ is a $C^\infty(M)$-**module**, not a vector space, and need not have a basis (the sphere cannot be combed). Local bases $\partial/\partial x^i$ exist on chart domains.
- A **tensor field** of type (r,s) is a $C^\infty(M)$-multilinear map $\Gamma(T^*M)^r \times \Gamma(TM)^s \to C^\infty(M)$. The gradient $\mathrm{d}f$ is a $(0,1)$ tensor field.